MILITARY SPECIFICATION

AMPLIFIER-CONTROL GROUP, AN/SIA-115

1. SCOPE

1.1 This specification covers the detailed requirements for an audio frequency amplifier-control group, type AN/SIA-115, for general announcing system, circuit 1MC.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-62 - Capacitors, Fixed, Electrolytic (DC Aluminum, Dry Electrolytic, Polarized.)
MIL-I-983 - Interior Communication Equipment, Naval Shipboard, Basic Design Requirements for.
MIL-C-3965 - Capacitors, Fixed, Nonsolid Electrolytic, (Tantalum,. Foil and Sintered -Slug) General Specification for.
MIL-A-20222 - Announcing Equipment, Shipboard.
MIL-A-21577 - Amplifiers, Audio Frequency, and Amplifier-Control Groups – Shipboard Announcing.

STANDARDS

MILITARY

MIL-STD-200 - Electron Tubes, Selection and Use of.
MIL-STD-275 - Printed Wiring for Electronic Equipment.
MIL-STD-701 - Preferred and Guidance Lists for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 The amplifier-control group AN/SIA-115 shall be in accordance with the applicable paragraphs of MIL-A-21577 and as specified herein. Whenever a requirement of MIL-A-21577 conflicts with a requirement of this specification, the requirement of this specification shall govern.

3.2 Equipment complement. - The equipment shall consist of eight power amplifiers, two amplifier oscillator modules, one relay section, and one test section in a common enclosure.

3.2.1 Preamplifiers shall be included as a part of each power amplifier.

3.2.2 An amplifier oscillator module shall consist of the plug-in type cards required to generate alarm signals in accordance with 3.19, a power supply common to all oscillator cards, and necessary wiring to connect the module to the system. Oscillator cards shall be provided with a device to prevent inadvertent insertion of cards in the wrong order of priority. The device shall allow intentional change in alarm priority by a simple adjustment.

3.3 Equipment function. - The function of the amplifier-control group, AN/SIA-115, described herein is to provide the control switching, amplification of voice and alarm signals, and generation of alarm signals required for general announcing system, circuit 1MC.

3.3.1 Circuit 1MC function. - The function of the general announcing system, circuit 1MC, is to provide one way transmission of orders and electronically generated alarm signals to all areas of a ship in which ship's personnel are located.

FSC 5830

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3.3.2 Microphone stations and loudspeakers. - Microphone stations and loudspeakers used in circuit 1MC shall be in accordance with MIL-A-20222.

3.3.3 Alarm contact makers. - All contact makers are normally open single pole switches. Contact makers for all alarms except the general alarm are held closed throughout the time the alarm is sounded. The contact maker for the general alarm, when actuated, momentarily closes the circuit and then opens.

3.4 System operation. - The amplifier-control group, AN/SIA-115, shall be designed to provide the types of operation specified in 3.4.1 through 3.4.4.

3.4.1 Microphone press-to-talk switches circuit 1MC. - The operation of a 1MC microphone press-to-talk switch shall:

(a) Connect the microphone audio circuit at that microphone station to the input of the amplifiers.

(b) Actuate appropriate system priority circuits.

(c) Place the amplifier in "ready" condition.

(d) Disconnect the adjacent loudspeaker on the local loudspeaker circuit associated with the microphone station box.

(e) Attenuate the ship's entertainment system.

3.4.2 Local loudspeaker cutout circuit. - A cutout circuit shall be included for each microphone station to automatically disconnect local loudspeakers when the microphone is in use. The cutout circuit shall consist of a normally closed relay which shall operate to open the audio circuit supplying the local loudspeakers when the microphone press-to-talk switch is operated.

3.4.3 Circuit SE attenuation. - Operation of any 1MC press-to-talk switch shall operate a normally open relay which shall complete an external attenuation circuit in the ship's entertainment system.

3.4.4 Circuit SE muting. - Operation of any alarm contact maker shall operate a relay which, through an external circuit, shall mute the output of the ship's entertainment system.

3.5 Enclosure. - The interior of the enclosure shall be divided into sections. The sections shall be further subdivided to accommodate various plug-in modules. Each section shall contain functional facilities as follows:

(a) Top section. -

(1) Forced air cooling fan or fans and ventilation louvers.
(2) Necessary terminal boards for interconnection between the cabinet and the ship's wiring.
(3) A front plate for access to the terminal boards.
(4) A top plate for terminal tubes.

(b) Control section. -

(1) A hinge mounted panel with safety stops to hold the panel in open position for servicing.
(2) Microphone and speaker disconnect switches, power supply switches, amplifier and oscillator transfer switches, fuses and power available indicator lamp mounted on the panel.

(c) Amplifier and oscillator section. -

(1) Subdivisions to accommodate drawer type power amplifier modules and amplifier oscillator modules mounted on guide rails.
(2) Connectors to provide connection of modules to the system.
(3) A safety catch on the right hand side of the module to prevent the inadvertent removal of the module from the cabinet.
(4) Provision to intentionally remove a module from the cabinet after releasing the safety catch.
(5) Dowels and matching bushings to provide support and alignment of amplifier modules and oscillator modules. Dowels and bushings shall be arranged to permit insertion of duplicate units only where an identical unit has been removed.

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(d) Relay and test section. -

(1) Unit construction similar to that for the amplifier and oscillator section shall be considered. If unit construction is impractical, the components shall be mounted on hinged panels.

3.6 Size. - Maximum overall dimensions of the amplifier-oscillator control rack shall be 69 inches in height, 20.5 inches in width and 15 inches in depth. Minimum practical dimensions consistent with design requirements will be acceptable.

3.7 Semiconductors. - Semiconductor circuits shall be used exclusively. Semiconductors and rectifiers shall be approved types listed in MIL-STD-701 and MIL-STD-200.

3.8 Capacitors. - Tantalum electroytic capacitors conforming to MIL-C-3965 shall be used in preamplifiers and power amplifiers. Plug-in-type electrolytics in accordance with MIL-C-62 may be used in the power supply.

3.9 Relays. - Relays, with the exception of power type contactors, may be hermetically sealed.

3.10 Printed wiring. - Printed wiring shall be in accordance with type C lead terminations specified in MIL-STD-275.

3.11 Wiring. - Wiring employed shall have insulation type T as shown in the applicable table of MIL-I-983.

3.12 Output power. - The rated output power for the individual power amplifier units shall be 20 watts, minimum rating.

3.13 Time constant. - Under the conditions specified in the overload limiter characteristic requirements of MIL-A-21577, the preamplifier output voltage shall return to the value specified in the overload limiter characteristic requirements within 0.025 second after the sudden increase in input signal voltage above reference level.

3.14 Interstage input facilities. - Interstage input circuit, input impedance and input level will not be required.

3.15 Power supply. - Each amplifier or amplifier oscillator unit shall be provided with its own power supply for all power requirements of the unit.

3.16 Output level. - The amplifier-oscillators shall be capable of maintaining an output level equal to that selected for the microphone preamplifiers. This voltage shall be used as a test signal source as specified in MIL-A-21577.

3.17 Adjustment of output level. - A screw-driver type adjustment with shaft locking shall be provided in the amplifier oscillator or in combination with the associated control circuits to permit independent adjustment of the amplifier oscillator output level for the alarm signal into the amplifier input.

3.18 Visual alarm. - Operation of alarm signals number 1 and 5 shall operate a relay for control of external visual alarm signal.

3.19 Alarm signals. - The following alarms shall be provided and shall have the following order of priority:

(a) Collision alarm.

(b) Chemical alarm.

(c) General alarm.

(d) Flight crash alarm.

(e) Unassigned.

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3.20 Input facilities. - The number and type of microphone control stations associated with the amplifier system shall be as follows:

Microphone station number	Microphone station type	Circuit controlled
1	IC/MSB-2	1MC
2	IC/MSB-2	1MC
3	IC/MSB-2	1MC
4	IC/MSB-2	1MC

3.21 Output facilities. - The number and designation of loudspeaker groups associated with the amplifier system shall be as follows:

Loudspeaker group number	Loudspeaker group designation	Circuit
1	Officers	1MC
2	Topside	1MC
3	Topside	1MC
4	Crew	1MC
5	Engineering spaces	1MC
6	Engineering spaces	1MC

3.22 Active circuit. - The active audio circuit shall consist of six power amplifiers. The remaining two amplifiers shall be considered as spares. They shall be used as replacement modules in the event of failure of active amplifiers. A selector switch shall permit selection of either oscillator module as the active module. The remaining oscillator module shall be used as the active module in the event of failure of the "active" module.

3.22.1 There shall be one loudspeaker group relay for each amplifier. Provision shall be made to operate group relays in parallel if one loudspeaker group requires the output from more than one amplifier.

3.23 Power supply switches. - Separate power switches shall be provided as follows:

Each amplifier unit.
Each amplifier-oscillator unit.
Relay unit.
Forced air cooling blower.

3.24 Test facilities. - Facilities shall be provided for the purpose of checking the operation of any of the amplifier modules or oscillator modules as an aid to system servicing and maintenance.

3.24.1 A voltmeter shall be provided to measure the output of a power amplifier or an oscillator module. The meter shall indicate zero db. for normal output.

3.24.2 A switch shall be provided to permit measuring the output of any amplifier selected for test. Selection of an amplifier for test shall start the chemical alarm oscillator of the inactive oscillator module for use as a test signal input to the amplifier.

3.24.3 A switch shall be provided to select and start any of the oscillators in the inactive module for use as a test signal to measure output of the oscillator module.

3.24.4 Testing shall be performed with the amplifiers in active service. Priorities shall be established to prevent test procedures from interfering with normal system operation. Design of the test facility shall prevent transmission of test signals over loudspeakers when checking amplifier or oscillator module operation.

3.24.5 A test socket, indicating lamps, and a test switch to test plug-in type relays shall be provided. Capability to test a relay for operating coil continuity, and for contact operation in both normal and operated conditions shall be provided.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling, examination and tests shall be as specified in MIL-A-21577.

5. PREPARATION FOR DELIVERY.

5.1 Preparation for delivery shall be as specified in MIL-A-21577.

6 NOTES

6.1 The notes specified in MIL-A-21577 are applicable to this specification.

Preparing activity:
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